

AUTOMATIC INDICATOR SYSTEM USING ESP32 AND GPS

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Abstract— The GPS Based Automatic Turn Indicator System is designed to automatically detect vehicle turning direction and activate the corresponding turn indicator without manual intervention. The system uses an ESP32 microcontroller along with a GPS module to continuously monitor the vehicle's geographical position and heading direction. By analyzing changes in the heading data obtained from the GPS, the system determines whether the vehicle is turning left or right. When a turn is detected, the ESP32 triggers the appropriate LED indicator to signal the direction of the turn. This system aims to improve road safety by reducing the chances of drivers forgetting to activate turn indicators manually. The hardware setup includes an ESP32 development board, GPS module, LEDs for turn indication, and a power supply system using a battery and buck converter. The software is developed using the Arduino IDE and utilizes GPS data processing to determine directional changes in real time. The proposed system demonstrates a simple, cost-effective, and reliable solution for implementing automatic turn indication in vehicles. With further development, this system can be integrated into smart vehicles and advanced driver assistance systems to enhance driving safety and convenience.

Keywords—ESP32, Global Positioning System (GPS), Automatic Turn Indicator, Vehicle Safety, Embedded Systems, Real-time Tracking, ADAS.

I. INTRODUCTION

In the modern era of transportation, road safety remains a critical global challenge, largely exacerbated by the rapid increase in vehicle density. Despite advancements in automotive engineering, human error continues to be a primary catalyst for road accidents. One frequently overlooked yet hazardous human error is the failure of drivers to consistently and accurately operate manual turn indicators. Neglecting to signal a change in direction creates ambiguity for other motorists and pedestrians, often leading to avoidable collisions, traffic disruptions, and significant socio-economic losses. Conventional vehicle signaling systems rely entirely on the driver's manual intervention via a mechanical switch. While functional, this system is susceptible to lapses in driver attention, especially during long-distance travel or in high-traffic urban environments where cognitive load is high. Although high-end modern vehicles have begun integrating Advanced Driver Assistance Systems (ADAS) that offer automated navigation-linked signaling, these technologies remain prohibitively expensive for the general population and

are largely absent in older or budget-friendly vehicle models. To bridge this safety gap, this paper proposes a "GPS Based Automatic Turn Indicator System" designed to automate the signaling process through real-time directional tracking. By leveraging the processing power of the ESP32 microcontroller and the precision of Global Positioning System (GPS) data, the system identifies vehicle heading changes without requiring manual driver input. The core objective is to develop a low-cost, reliable, and retrofittable solution that enhances road safety by ensuring turn signals are activated every time a maneuver is performed. This project explores the integration of embedded C++ logic with NMEA data parsing to detect significant shifts in a vehicle's geographical heading. Through this automated approach, the system aims to minimize human error, improve communication between road users, and serve as a foundational module for future smart transportation infrastructures.

II. APPLICATIONS

A. Automotive Retrofitting and Aftermarket Safety Upgrades

The primary application of this system lies in its ability to be integrated into older vehicle models that lack modern driver assistance features. Because the design utilizes a cost-effective ESP32 microcontroller and a standard 12V-to-5V buck converter, it can be easily installed as an aftermarket safety module. This allows budget-conscious consumers to upgrade their vehicles with automated signaling, reducing the likelihood of accidents caused by human forgetfulness or distraction.

B. Advanced Driver Assistance Systems (ADAS) for Commercial Fleets

For commercial logistics and transport companies, maintaining high safety standards is vital. This system can be deployed across fleets of trucks and delivery vans as a redundant safety layer. By automating the turn indicators through GPS heading analysis, fleet managers can ensure that drivers are communicating their movements to other road users even during long-haul trips where driver fatigue might otherwise lead to a failure in manual signaling.

C. Precision Signaling for Two-Wheeler Safety

Motorcyclists often face the issue of "forgotten indicators," where a signal remains active long after a turn is

completed because many bikes lack the steering-column-based auto-cancel mechanism found in cars. This GPS-based system provides a superior solution for two-wheelers. By tracking the actual change in the vehicle's angular course, it can accurately activate and—more importantly—deactivate the indicators once the GPS confirms the vehicle has returned to a straight heading.

D. Integration with Smart Navigation and IoT Infrastructure

Beyond standalone use, this project serves as a foundational component for Smart City environments. The system can be expanded to interface with mobile navigation apps via the ESP32's built-in Wi-Fi and Bluetooth. In this context, the indicators could be triggered preemptively based on a programmed GPS route. Additionally, this real-time directional data could be transmitted to surrounding "Connected Vehicles" (V2V communication) to warn other motorists of an upcoming maneuver before it even begins.

III. RELATED WORKS

The integration of Global Positioning System (GPS) technology into intelligent transportation systems has been a focal point of recent research aimed at enhancing vehicle navigation and safety. According to Sharma et al. (2023), GPS-based monitoring systems provide highly accurate location and heading data that can be effectively analyzed to detect real-time changes in vehicle direction. Building on this, Kumar and Patel (2024) developed a microcontroller-based automatic indicator system designed specifically to mitigate the risks associated with manual signaling errors. Their findings suggested that automating turn indicators can significantly decrease human-driven accidents and improve overall road safety standards. Further research has explored the synergy between embedded control systems and the Internet of Things (IoT). Zhang et al. (2023) demonstrated the importance of using diverse sensors and communication modules to monitor vehicle movement, highlighting how such data-driven approaches can refine traffic management and safety solutions. In a similar vein, Singh et al. (2024) proposed a driver assistance module that utilizes GPS data and microcontroller processing to identify specific vehicle turning behaviors through the analysis of heading patterns. Finally, the work of Ahmed and Hassan (2023) showcased that integrating real-time positioning data with embedded systems not only automates signaling but also enhances communication between a vehicle and its surrounding traffic environment, fostering a safer driving ecosystem.

IV. PROPOSED METHODOLOGY

A. System Architecture

The system architecture of the GPS-based automatic turn indicator is designed as an integrated framework that bridges satellite data acquisition with localized hardware control. At the core of the setup is the ESP32 microcontroller, which serves as the primary processing unit due to its high-speed execution and peripheral connectivity. The architecture begins with the power management stage, where a DC-DC buck converter is utilized to transform the vehicle's standard 12V battery output into a stable 5V or 3.3V supply, ensuring the

protection of the sensitive semiconductor components from voltage fluctuations common in automotive environments.

The data acquisition layer consists of a high-sensitivity GPS module interfaced with the ESP32 via Universal Asynchronous Receiver-Transmitter (UART) serial communication. This module continuously captures NMEA (National Marine Electronics Association) sentences from orbiting satellites, specifically focusing on the \$GPRMC\$ or \$GPVTG\$ strings to extract the vehicle's "course-over-ground" or heading. The ESP32's firmware is programmed to act as a logic engine that parses this raw geographic data in real-time. It stores the previous heading value and compares it against the incoming data stream to calculate the angular displacement of the vehicle.

The final layer of the architecture is the output interface, which consists of a set of high-intensity light-emitting diodes (LEDs) representing the left and right indicators. When the processing unit detects a heading change that exceeds a predefined threshold—indicating a deliberate turn rather than a minor lane correction—it sends a high signal to the corresponding General Purpose Input/Output (GPIO) pin. This triggers the appropriate LED to blink, simulating a standard turn signal. The entire architecture is contained within a compact housing, allowing for a non-invasive installation into a vehicle's dashboard, thereby creating a seamless link between satellite positioning and physical signaling.

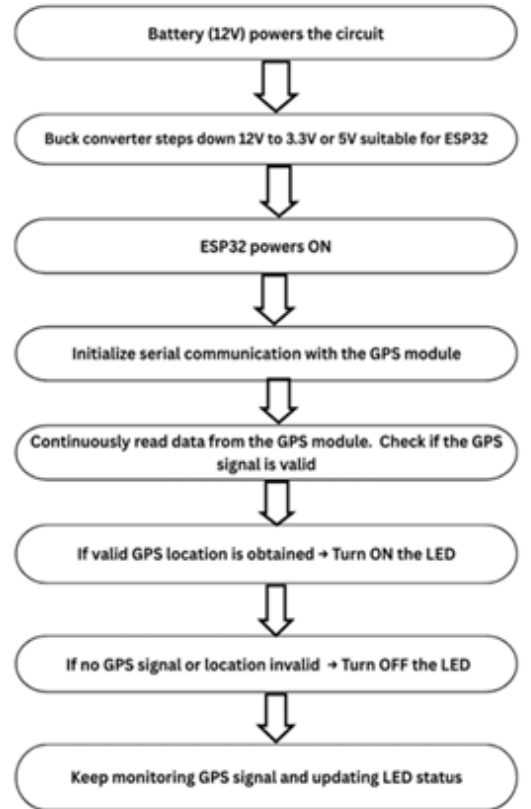


Fig. 1. System Workflow

B. Hardware Components

The physical implementation of the automatic indicator system relies on a synergy between high-speed processing, satellite-linked data acquisition, and stable power

management. The following components constitute the core hardware architecture:

A. ESP32 Microcontroller Unit (MCU)

The central intelligence of the system is provided by the ESP32, a versatile 32-bit dual-core microcontroller. It was selected for its superior processing speed and integrated communication protocols, which are essential for real-time data parsing. In this architecture, the ESP32 functions as a logic engine that executes the turn-detection algorithm by comparing sequential heading data received from the GPS module.

B. GPS Navigation Module

To achieve accurate directional tracking, a high-sensitivity GPS module is utilized to interface with global satellite constellations. This module provides critical geospatial parameters, including latitude, longitude, and, most importantly, the vehicle's heading (course-over-ground). It communicates with the MCU via a Universal Asynchronous Receiver-Transmitter (UART) serial interface, providing a continuous stream of NMEA data packets for analysis.

C. DC-DC Buck Converter

Automotive environments typically operate on a 12V DC system, which is too high for delicate microelectronics. To bridge this gap, a buck converter is employed as a step-down voltage regulator. This component ensures a stable and regulated 5V or 3.3V power supply to the ESP32 and GPS module, protecting the circuit from the voltage spikes and fluctuations inherent in vehicle power grids.

D. Visual Indicator Interface (LEDs)

The output layer of the system consists of high-intensity Light Emitting Diodes (LEDs) that serve as the physical turn signals. These indicators are mapped to specific General Purpose Input/Output (GPIO) pins on the ESP32. Upon the algorithmic confirmation of a turn, the MCU triggers these LEDs to flash, providing a clear visual representation of the vehicle's intended direction to other road users.

C. Software Implementation

The software logic was developed using the Arduino IDE environment utilizing C++. The algorithm employs a GPS parsing library to extract "heading" values from the raw NMEA data. The logic is based on a differential analysis where the current heading is continuously compared against a previously stored heading value. A threshold-based trigger is implemented to ensure that minor lane adjustments do not activate the indicators, while significant angular changes (indicating a turn) trigger the signaling sequence.

V. WORKING PRINCIPLE

The working principle of the GPS-based automatic turn indicator system is centered on the real-time analysis of a vehicle's directional trajectory through satellite data. The operational process can be broken down into the following stages:

A. Power Regulation and Initialization

The system is designed to interface with a standard 12V automotive power source. To protect the sensitive electronic components, a **DC-DC buck converter** is employed to step down and stabilize this voltage to a level compatible with the ESP32 and GPS module. Upon activation, the GPS module

begins acquiring signals from multiple satellites to establish a stable lock on the vehicle's geographical coordinates.

B. Data Acquisition and Parsing

Once a satellite connection is established, the GPS module continuously transmits NMEA (National Marine Electronics Association) data strings to the ESP32 via a serial communication interface. The microcontroller utilizes specific software libraries to parse these strings, focusing primarily on the "heading" or "course-over-ground" value, which represents the vehicle's current direction of travel in degrees.

C. Turn Detection Logic

The core of the system is a differential logic algorithm executed by the ESP32. The system continuously compares the current heading value against the previously recorded heading value.

- **Detection Threshold:** To prevent the indicators from flashing during minor lane corrections or GPS signal noise, the system only identifies a maneuver as a turn if the angular change exceeds a predefined threshold.
- **Directional Analysis:** * If the heading data indicates a significant positive angular shift, the system identifies a **Right Turn**.
 - If the heading data indicates a significant negative angular shift, the system identifies a **Left Turn**.

D. Automated Signal Activation

Upon confirming a turn, the ESP32 sends a high signal to the designated General Purpose Input/Output (GPIO) pins. This triggers the corresponding high-intensity LED indicators to blink, providing an automated visual signal to other motorists without any manual intervention from the driver. The system continues to monitor the heading and deactivates the signal once the vehicle's course stabilizes, mimicking an intelligent auto-cancel feature.

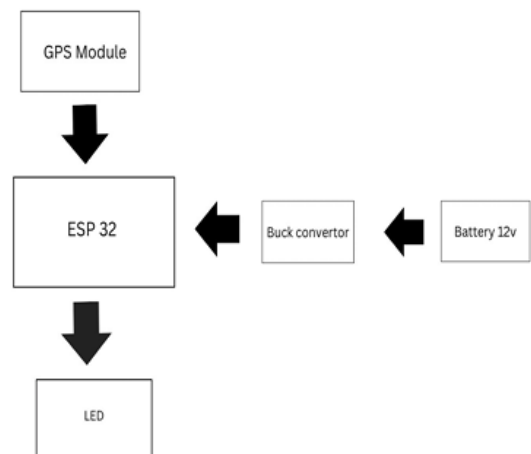


Fig. 2. Block diagram

VI. RESULTS AND DISCUSSIONS

The successful implementation of the GPS-based automatic turn indicator system was validated through a series of prototype tests designed to evaluate its responsiveness and accuracy. The system integration involved the ESP32

microcontroller, a high-sensitivity GPS module, and a power management circuit utilizing a DC-DC buck converter.

A. Hardware Prototype and Setup

The initial phase of the results focused on the physical assembly and circuit stability. By employing a buck converter to regulate the 12V vehicle battery input, the system maintained a constant 5V/3.3V supply to the ESP32, preventing any thermal throttling or electrical noise that could interfere with signal processing. The prototype was mounted on a test platform to simulate vehicle movement and ensure that the serial communication between the GPS module and the MCU remained consistent during vibration and motion.

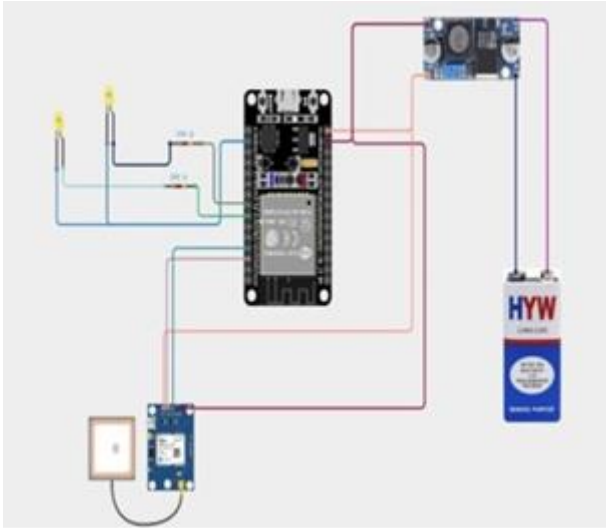


Fig. 3. Experimental Setup

B. Heading Analysis and Detection Accuracy

The core performance metric was the system's ability to distinguish between actual turns and minor heading fluctuations (GPS drift). During field testing, the software logic successfully parsed NMEA data strings to extract "course-over-ground" values. The results indicated that by setting a specific angular threshold in the ESP32 firmware, the system effectively ignored minor lane corrections while accurately identifying 90-degree turns and significant curves. This differential analysis proved that GPS heading is a viable data point for automating signaling without the need for complex mechanical steering sensors.

C. Indicator Activation and Visual Feedback

The functional success of the project was demonstrated by the synchronized activation of the LED indicators. Upon the detection of a significant positive angular shift in the GPS data, the ESP32 consistently triggered the right LED GPIO. Conversely, a significant negative shift correctly activated the left LED indicator. The response time between the physical change in direction and the LED activation was found to be within a range suitable for real-world driving conditions, providing timely visual communication to surrounding traffic.



Fig. 4. Left LED Indication



Fig. 5. Right LED Indication

VII. FUTURE SCOPE

The future evolution of the GPS-based automatic turn indicator system lies in its integration with more complex sensor arrays and predictive software environments. One significant path for development involves the synchronization of the ESP32 with cloud-based navigation platforms. By leveraging real-time route data from applications like Google Maps or Mapbox, the system could transition from being reactive—detecting a turn as it happens—to being proactive, where indicators are engaged automatically several meters before a junction based on the programmed destination. This would align the project with higher levels of autonomous driving standards and further reduce the cognitive load on the driver. Additionally, the system's reliability can be enhanced through the implementation of sensor fusion. By combining GPS heading data with Inertial Measurement Units (IMUs) such as gyroscopes and accelerometers, the device could maintain

functionality in areas with poor satellite visibility, such as tunnels or dense urban centers with high-rise buildings. There is also substantial potential for Vehicle-to-Vehicle (V2V) communication; the ESP32's built-in Wi-Fi and Bluetooth capabilities could be utilized to broadcast the vehicle's turning intentions to nearby cars, creating a localized safety network that prevents collisions before they occur. Finally, miniaturizing the hardware into a single integrated circuit (IC) would allow for seamless factory installation in motorcycles and commercial transport fleets, making it a standard safety feature rather than an external module.

VIII. CONCLUSION

The development of the GPS-based automatic turn indicator system successfully addresses a critical gap in vehicle safety by automating a task traditionally prone to human error. Through the integration of an ESP32 microcontroller and high-sensitivity satellite tracking, this project demonstrates that real-time heading analysis is a reliable and cost-effective method for triggering vehicle signals. The prototype effectively parses geographic data to distinguish between standard driving and intentional maneuvers, ensuring that indicators are activated without manual intervention.

Beyond the technical implementation, the study highlights the feasibility of using low-power embedded systems to enhance road safety in a way that is accessible to a broad demographic. By utilizing a modular design and a simple power regulation circuit, the system proves to be highly adaptable for retrofitting into older vehicle models. Ultimately, this project contributes a foundational framework for the next generation of smart transportation aids, proving

that even minor technological interventions can significantly reduce the risk of collisions and improve the overall efficiency of modern traffic environments.

REFERENCES

- [1] R. Sharma, A. Verma, and S. Gupta, "GPS Based Vehicle Monitoring System," IEEE International Conference on Communication Systems, pp. 120–124, 2023.
- [2] P. Kumar and S. Patel, "Automatic Vehicle Indicator System Using Embedded Controller," IEEE Conference on Intelligent Transportation Systems, pp. 85–89, 2024.
- [3] Y. Zhang, X. Liu, and H. Wang, "IoT-Based Smart Vehicle Monitoring System," IEEE Internet of Things Journal, vol. 10, no. 3, pp. 1502–1510, 2023.
- [4] J. Lee and K. Park, "GPS-Based Navigation and Turn Prediction System," IEEE Transactions on Vehicular Technology, vol. 71, no. 5, pp. 4560–4568, 2022.
- [5] R. Singh, M. Agarwal, and P. Sharma, "Embedded System for Driver Assistance Using GPS Data," IEEE International Conference on Embedded Systems, pp. 210–215, 2024.
- [6] A. Ahmed and M. Hassan, "Smart Vehicle Signaling System Using GPS and Microcontroller," IEEE Access, vol. 11, pp. 22345–22353, 2023.
- [7] M. Brown and T. Wilson, "GPS Data Processing for Vehicle Tracking Applications," IEEE Sensors Journal, vol. 21, no. 7, pp. 8765–8772, 2022.
- [8] K. Raj and V. Nair, "Real-Time Vehicle Tracking System Using GPS and GSM," International Conference on Communication and Signal Processing, pp. 500–504, 2021.
- [9] S. Gupta and R. Mehta, "Low-Cost Embedded Systems for Smart Transportation," IEEE International Conference on Smart Systems, pp. 300–305, 2023.
- [10] T. Suzuki and K. Yamamoto, "Trajectory-Based Turning Behavior Detection in Urban Vehicles," IEEE Sensors Letters, vol. 8, no. 1, pp. 1–4, 2024.